

Ekansh Tardeja

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Gender: Male **Date of birth:** 28/10/2007 **Place of birth:** **Nationality:** Indian

ABOUT ME

I am an undergraduate Physics student with a strong interest in observational astrophysics research. I actively engage in astrophotography as a means of conducting systematic observations and analyzing the data obtained from them, with particular interest in photometric studies. In addition, I participate in science outreach initiatives aimed at communicating astronomical concepts clearly and effectively to a broader audience. I am also an active member of the Astronomy Club at Fergusson College (Autonomous).

EDUCATION AND TRAINING

[07/2025 – Current]

Bachelor of Science in Physics

Fergusson College (Autonomous) www.fergusson.edu

City: Pune | **Country:** India | **Field(s) of study:** Natural sciences, mathematics and statistics: ● *Physics* ● *Mathematics*

-Current CGPA: 8.95

-Member of Astro Club

-As an active member of the Astro Club, I contributed to our annual science outreach poster exhibition, where we communicate physics concepts to the general public through interactive models, posters, and demonstrations. I also volunteered in organizing our annual national-level seminar, which brings together professors and researchers from leading universities to deliver talks and engage with students.

PROJECTS

[02/2026 – 03/2026]

Times of minima calculator from lightcurves

Python, Scipy, Astropy, Matplotlib

https://github.com/ekanshtardeja/TimesOfMinimaPipeline.git

1. Expects the name of an eclipsing binary star.
2. The pipeline searches MAST for available light curves of the target.
3. Retrieved light curves are processed and plotted.
4. Gaussian functions are fitted to identify eclipse minima.
5. Times of minima are calculated from the fitted profiles.

This project was done under the guidance of a PHD student at Osmania University, India (Ravi Raja Pothuneni).

Goal was to do a period variation analysis of 2 Over Contact Eclipsing Binaries(V2284 Cygni and V1918 Cygni).

[08/2026 – 09/2026]

Automated RV calculation from star spectras

Python, Scipy, Astropy

https://github.com/ekanshtardeja/RVsFromSpectra.git

1. Expects the name of a star.
2. The pipeline uses astroquery and queries spectrum data from ESO.
3. Retrieved spectra is plotted, normalized.

4. Gaussian functions are fitted to the given rest wavelengths to calculated observed wavlengths.
5. From these calculated wavelengths, Radial Velocity is calculated.

This project was done under the guidance of a masters student at LMU Munich (Jebraan Mudholkar).

Goal was to do an independent extraction of stellar radial velocities from publicly available WASP-18 spectra using a line-by-line Gaussian fitting approach, with the aim of recovering the periodic radial velocity signal induced by its known exoplanet.

SKILLS

Jupyter | Matplotlip | Numpy | Scipy | Astropy | python | MATLAB | Git | Javascript

VOLUNTEERING

[02/2024 – Current] **Science Outreach**

Aeronautics and Space Exploration (AXSX)

- Organizes stargazing and science outreach programs for the general public.
- https://axsx.in/

I work as a Science Communicator during the organization's stargazing sessions, where I engage with participants by explaining astronomical objects, observational techniques, and fundamental astronomy concepts through guided telescope observations and interactive discussions.

[01/02/2026 – 28/02/2026] **Science Outreach** Inter-University Centre for Astronomy and Astrophysics(IUCAA), Pune

Participated in the open day at IUCAA on National Science Day, where I made and explained a model of Nuclide Chart to the public. This is an annual event wherein people of all age groups and backgrounds visit and learn about science, and the activities and research that happen in IUCAA.

[31/07/2025 – Current] **Science Outreach** Fergusson College (Autonomous), Pune

Organized and partook in outreach events like the annual "Poster Exhibition" and annual two day national seminar "Frontiers in Physics" as a member of the Astro-Club in Fergusson College.

LANGUAGE SKILLS

Mother tongue(s): Hindi

Other language(s):

English

LISTENING C2 **READING** C2 **WRITING** C2

SPOKEN PRODUCTION C2 **SPOKEN INTERACTION** C2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user